14.2. Bipolar Junction Transistor Construction

A bipolar junction transistor has, essentially, three regions known as emitter, base and collector. All these three regions are provided with terminals, which are labelled as *E* (for emitter), *B* (for base) and *C* (for collector) respectively. A brief description of the above regions is discussed below:

1. ***Emitter.*** It is a region situated in one side of transistor, which supplies charge carries (*i.e.*, electrons or holes) to the other two regions. The emitter is a heavily doped region.
2. ***Base.*** It is the middle region that forms two P-N junctions in the transistor. The base of transistor is thin, as compared to the emitter and is a lightly doped region.
3. ***Collector.*** It is a region situated in the other side of transistor (*i.e.*, the side opposite to the emitter), which collects charge carriers (*i.e.*, electrons or holes). The collector of a transistor is always larger than the emitter and base of a transistor. The doping level of the collector is intermediate between the heavy doping of emitter and the light doping of the base.

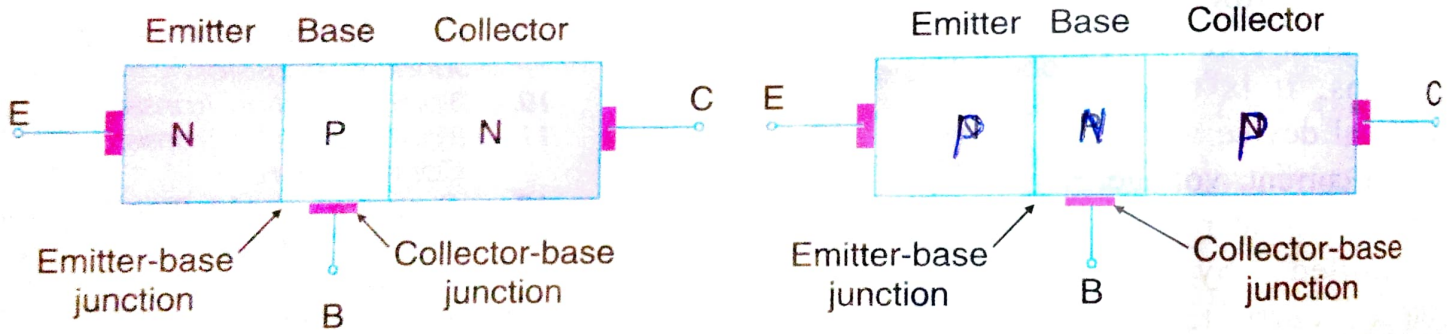
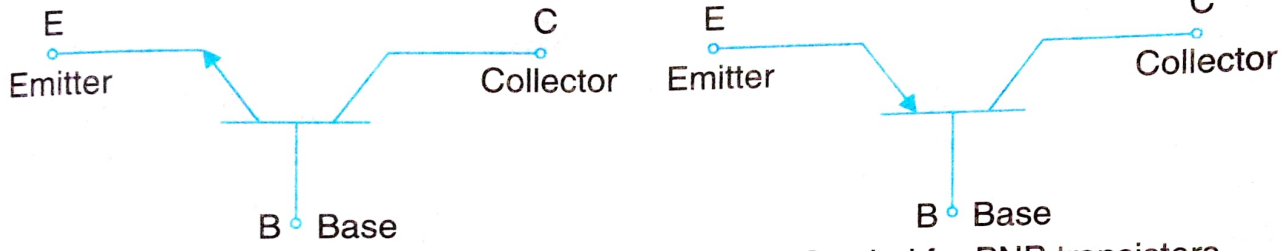


Fig. 14.2. Bipolar JunctionTransistor construction.

14.5. Bipolar Junction Transistor Symbols

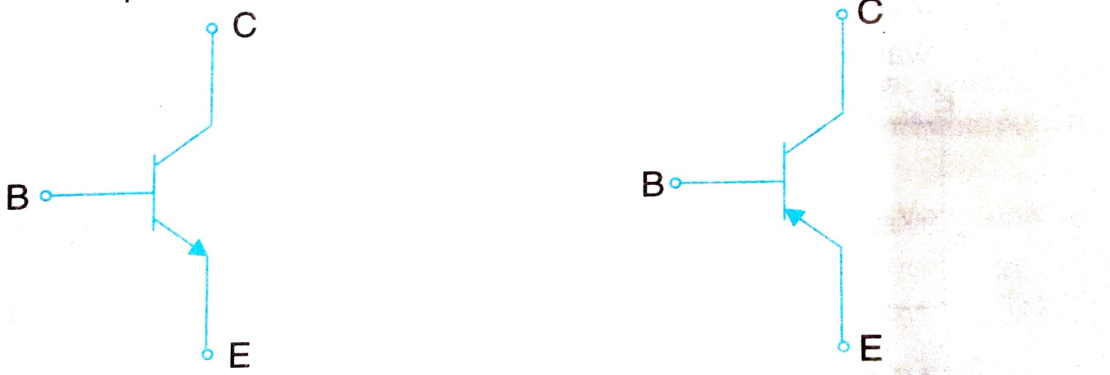
We have already mentioned in the Art. 14.2 that there are two types of transistors namely NPN and PNP. When the transistor is used as circuit element, it is difficult to represent it with Fig. 14.2 (*a*) and (*b*). In order to make it convenient, both the transistors are represented by their circuit symbols as shown in Fig. 14.6 (*a*) and (*b*) respectively. The transistor symbol carries an arrowhead in the emitter pointing from the P-region towards the N-region. The arrowhead indicates the direction of a conventional current flow in a transistor.



(*a*) Symbol for NPN transistors. (*b*) Symbol for PNP transistors.

Fig. 14.6. Bipolar Junction Transistor symbols.

It may be noted that the direction of arrowheads at the emitter in NPN and PNP transistors is opposite to each other. Sometimes, it is more convenient to represent the symbols for NPN and PNP transistors as shown in Fig. 14.7 (*a*) and (*b*).



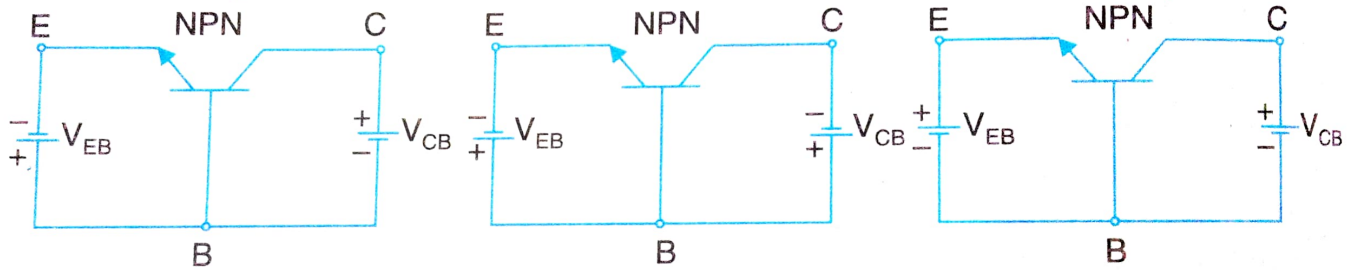
(*a*) Symbol for NPN transistor (*b*) Symbol for PNP transistor

Fig. 14.7. Unbaised transistor.

14.7. Bipolar Junction Transistor Biasing

The application of a suitable dc voltages, across the transistor terminals, is called biasing. Each junction of a transistor may be forward biased or reverse-biased independently. There are following three different ways of biasing a transistor, which are also known as modes of transistor operation.

1. ***Forward-active.*** In this mode, the emitter-base junction of a transistor is forward biased and the collector base junction is reverse biased as shown in Fig. 14.9 (*a*). In a forward active biasing the negative terminal of a battery is connected to N-side and positive terminal to P-side. The reverse biasing requires all the connections to be opposite to those for forward biasing.



(*a*) Forward-active. (*b*) Saturation. (*c*) Cut-off.

Fig. 14.9. Different modes of transistor biasing.

2. *Saturation.* In this mode, both the emitter-base and collector-base junctions of a transistors are forward-biased as shown in Fig. 14.9 (*b*). In this mode, the transistor has a very large value of current. The transistor is operated in this mode, when it is used as a closed switch.

3. *Cut-off. In this mode*, both the emitter-base and collector-base junctions of a transistor are reverse biased as shown in Fig. 14.9 (*c*). In this mode, the transistor has practically zero current. The transistor is operated in this mode, when it is used as an open switch.

The three different modes are summarized in Table 14.1. It must be remembered that for amplifier, a transistor is used in forward-active mode and for switching applications (*i.e.*, ON and OFF applications) a transistor is used alternatively is saturation and cut-off modes.

Table 14.1: Modes of transistor operation

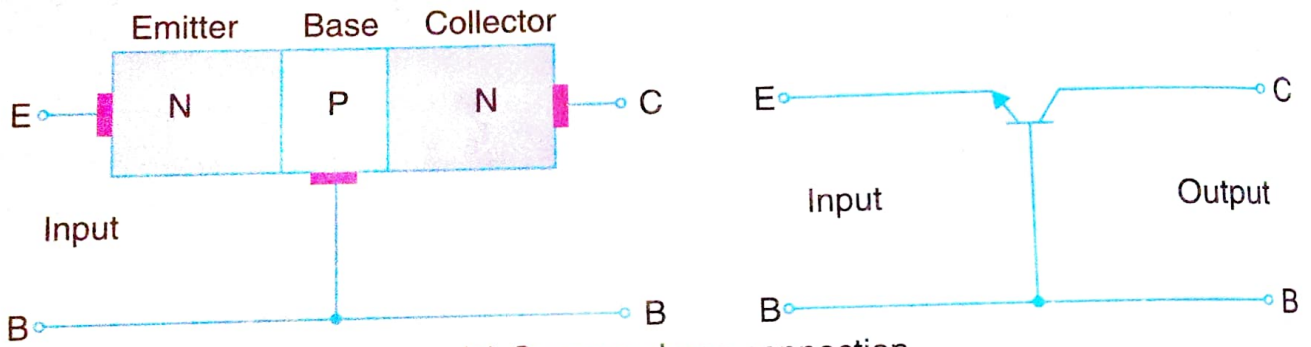
S. No.	Mode	Junction-bias condition	
		Emitter Base	Collector Base
1.	Forward active	Forward	Reverse
2.	Saturation	Forward	Forward
3.	Cut-off	Reverse	Reverse

Note. The modes of operation are the same for both NPN and PNP transistors. However, in PNP transistor biasing, the polarity of battery voltages are opposite to that of NPN transistor.

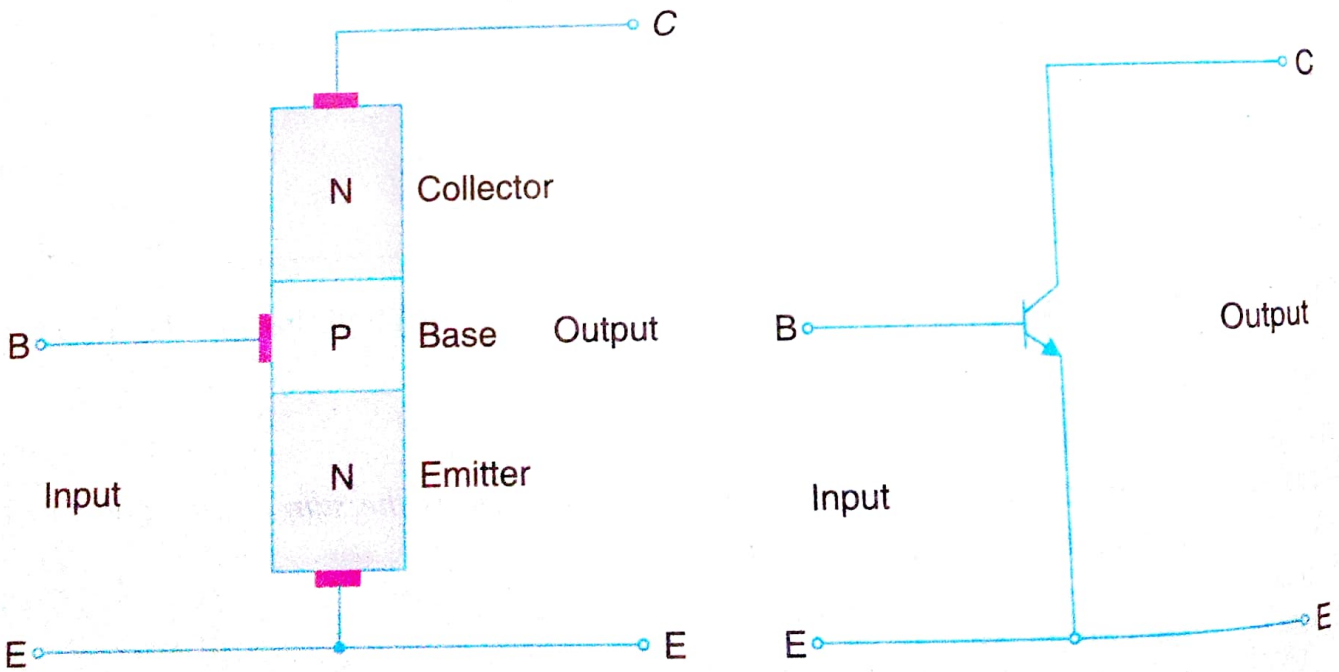
14.11. Bipolar Junction Transistor Circuit Configurations

We know that a transistor has three terminals or leads namely emitter (E), base (B) and collector (C). However, when a transistor is connected in a circuit, we require four terminals *i.e.*, two terminals for input and two for output. This difficulty is overcome by using one of the three terminals as a common terminal to the input and output terminals. Depending upon the terminals, which are used as a common terminals, the transistors can be connected in the following three different connections or configurations:

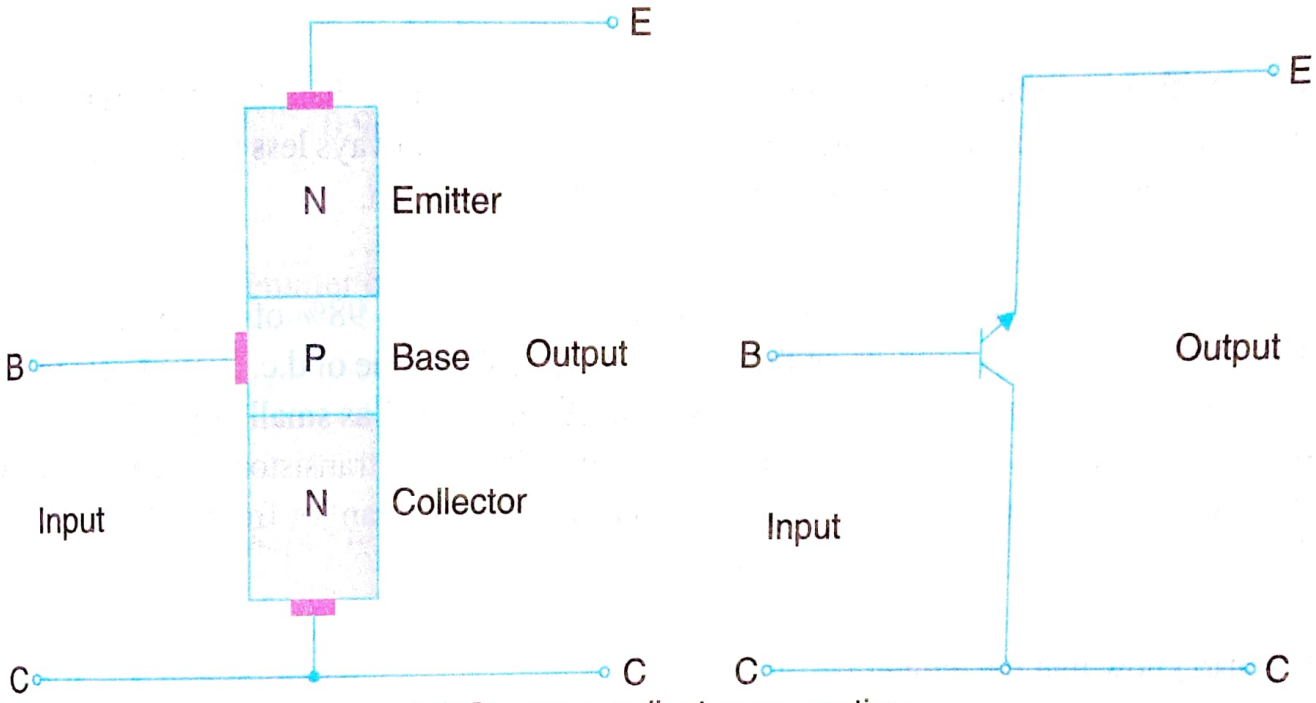
1. ***Common-base (CB) connection.*** In this configuration, the transistor is connected with the base as a common terminal as shown in Fig. 14.13 (a). The input is applied between the emitter and base terminals. The output is taken between the collector and base terminals. This type of configuration is used to explain the operation of NPN and PNP transistors.



(a) Common-base connection.



(b) Common-emitter connection.



(*c*) Common-collector connection.

Fig. 14.13. Transistor circuit connections or configurations.

2. ***Common-emitter (CE) connection.*** In this configuration, the transistor is connected with the emitter as a common terminal as shown in Fig. 14.13 (*b*). The input is applied between the base and collector terminals. The output is taken between collector and emitter terminals. This is one of the most commonly used connections of a transistor.
3. ***Common-collector (CC) connection.*** In this configuration, the transistor is connected with collector as a common terminal as shown in Fig. 14.13 (*c*). The input is applied between the base and collector terminals. The output is taken between emitter and collector terminals.

15.6. Characteristics of a Transistor in Common-Emitter Configuration

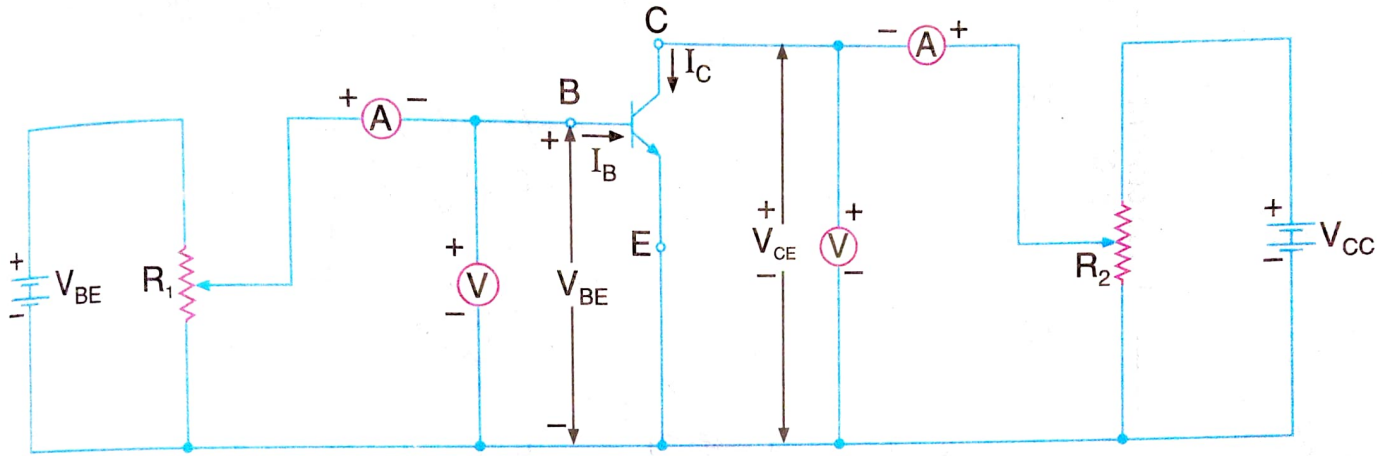


Fig. 15.4. Circuit arrangement for determining common-emitter transistor characteristics.

A transistor, in a common-emitter configuration, has also two important characteristics namely input characteristics and output characteristics as discussed below:

1. ***Input characteristics.*** These curves give the relationship between the base current (I_B) and the base-to-emitter voltage (V_{BE}) for a constant collector-to-emitter voltage (V_{CE}).
2. ***Output characteristics.*** These curves give the relationship between the collector current (I_C) and collect-to-emitter voltage (V_{CE}) for a constant base current (I_B).

These characteristics may be obtained by means of a circuit arrangement as shown in Fig. 15.4. In this circuit, an NPN transistor is connected in a common-emitter configuration. The collector-to-emitter voltage (V_{CE}) can be varied by changing the position of potentiometer , while the base-to-emitter voltage (V_{BE}) can be varied by changing the position of potentiometer R_1. The dc milliammeters and dc voltmeters are connected in the base and the collector circuit of a transistor to measure the voltages and currents.

15.7. Input Characteristics of a Transistor in Common-Emitter Configuration

These characteristics may be obtained by using the circuit arrangement shown in Fig. 15.4. First of all, we adjust the collector-to-emitter voltage (V_{CE}) to 1 volt. Then we increase to base-to-emitter voltage (V_{BE}) in small suitable steps and record the corresponding values of base current (I_B) at each step. If we plot a graph with base-to-emitter voltage (V_{BE}) along the horizontal axis and the base current (I_B) along the vertical axis, we shall obtain a curve marked $V_{BE} = 1$ V as shown in Fig. 15.5. A similar procedure may be used to obtain characteristics at different values of collector-to-emitter voltage *i.e.*, $V_{CE} = 2$, 10 and 20V. The input characteristic gives us the information about the following important points:

1. There exists a threshold or knee voltage (V_K) below which the base current is very small. The value of knee voltage is 0.5 V for silicon and 0.1 V for germanium transistors.

2. Beyond the knee, the base current (I_B) increases with the increase in base-to-emitter voltage (V_{BE}) for a constant collector-to-emitter voltage (V_{CE}). However, it may be noted that the value of base current does not increase as rapidly as that of the input characteristic of a common-base transistor. It means that input resistance of a transistor in common-emitter configuration is higher as compared to the common base configuration.
3. As the collector-to-emitter voltage (V_{CE}) is increased above 1 V, the curves shift downwards. It occurs because of the fact, that as V_{CE} is increased, the depletion width in the base-region increases. The reduces the effective base width, which in turn reduces the base current.

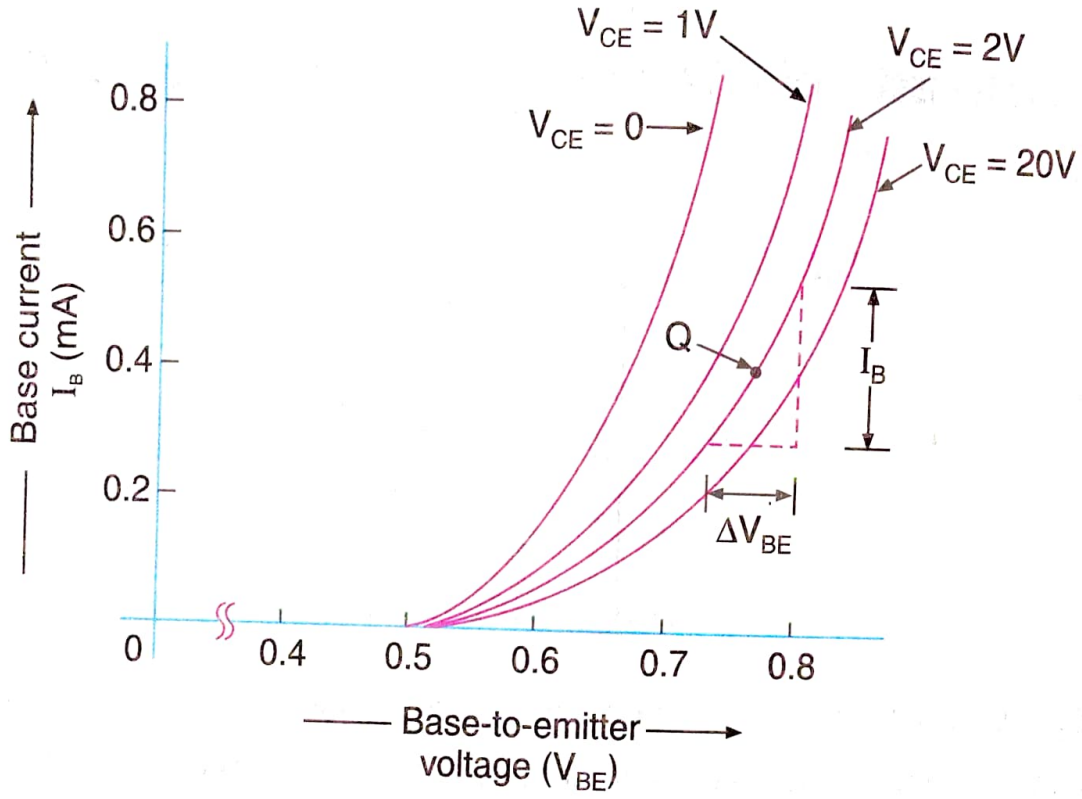


Fig. 15.5. Input characteristics of a common-emitter transistor.

4. The input characteristic may be used to determine the value of common-emitter transistor a.c. input resistance (R_i). Its value is given by the ratio of change in base-to-emitter voltage (ΔV_{BE}) to the resulting change in base current (I_B) at a constant collector-to-emitter voltage (V_{CE}). Mathematically, the a.c. input resistance,

$$R_i = \frac{\Delta V_{BE}}{\Delta I_B}$$

It may be noted that the input characteristic is not linear in the lower region of the curve. Therefore, the input resistance varies with the location of the operating point. The value of a.c. input resistance ranges from 600 Ω to 4000 Ω.

15.8. Output Characteristics of a Transistor in Common-Emitter Configuration

These characteristics may be obtained by using the circuit shown in Fig. 15.4. To begin with, adjust the base current (I_B) to 40 mA value. Then increase the collector-to-emitter voltage (V_{CE}) in a number of steps and record the corresponding values of collector current (I_C) at each step. If we plot a graph with collector-to-emitter voltage (V_{CE}) along the horizontal axis and collector current (I_C) along the vertical axis, we shall obtain a curve marked $I_B = 40$ mA as shown in Fig. 15.6. A similar procedure may be used to obtain characteristics at $I_B = 80$ mA , 120 mA, 160 mA and so on. The output characteristics give us the information about the following important points:

1. The output characteristics may be divided into three important regions namely saturation region, active region and cut-off region. The saturation and cut-off regions are shown by the

shaded areas, while the active region is the region between the saturation and cut-off region.

2. As the collector-to-emitter voltage (V_{CE}) is increased above zero, the collector current (I_C) increases rapidly to a saturation value, depending upon the value of base current. It may be noted that collector current (I_C) reaches to a saturation value when V_{CE} is about 1 V.
3. When collector-to-emitter voltage (V_{CE}) is increased further, the collector current slightly increases. This increase in collector current is due to the fact that increased value of collector-to-emitter voltage (V_{CE}) reduces the base current and hence the collector current increases. This phenomenon is called an early effect.
4. When the base current is zero, a small collector current exists. This is called leakage current. However, for all practical purposes, the collector current (I_C) is zero, when the base current (I_B) is zero. Under this condition the transistor is said to be cut-off.

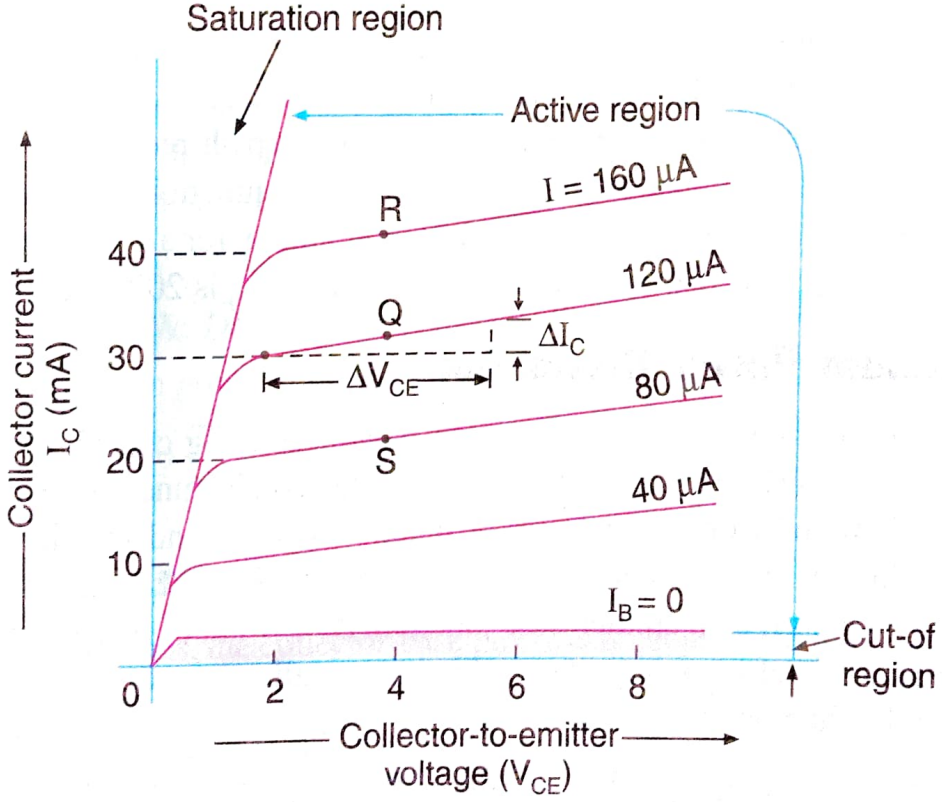


Fig. 15.6. Output characteristics of a common-emitter transistor.

5. The characteristic may be used to determine the common-emitter transistor a.c. output resistance. Its value at any given operating point Q is given by the ratio of a change in collector-to-emitter voltage (ΔV_{CE}) to the resulting change in collector current (ΔI_C) for a constant base current. Mathematically, the a.c. output resistance,

$$R_0 = \frac{\Delta V_{CE}}{\Delta I_C}$$

The common-emitter output resistance of a transistor range from 10 kΩ to 50 kΩ.

6. The characteristic may be used to determine the small-signal common-emitter current gain or a.c. beta (β_0) of a transistor. This can be done by selecting two points R and S on the characteristics and note down the corresponding valued of ΔI_C and ΔI_B. Thus if $\Delta I_C = 40 - 20 = 20\ \text{mA} = 20 \times 10^{-3}$ A and $\Delta I_B = 160 - 80 = 80\ \text{mA} = 80 \times 10^{-6}$ A then small signal common emitter current again,

$$\beta_0 = \frac{\Delta I_C}{\Delta I_B} = \frac{20 \times 10^{-3}}{80 \times 10^{-6}} = 250$$